

Supplementary material: Signed Network Dataset Repository: Extracting Signed Relations from Social Networks

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1 DATASETS

1.1 Reddit Posts Dataset

This dataset is collected from Reddit¹ social media platform. Reddit's vast scaled discussion board is very much like other social media platforms. Reddit socially aggregates millions of individuals who interact with each other by sharing videos, images, news, or text regularly. Reddit divides the entire platform into several designated subreddits, each designated for discussion on similar topics or socially aggregated communities sharing the same interests (e.g., posts with videos and posts related to similar topics). Within each subreddit, users share their views via images, texts, and videos which are referred to as Reddit *posts*. These Reddit *posts* are further discussed by other users using dedicated *comment threads*. Furthermore, each Reddit *post* or *comment* can be rewarded upvotes or downvotes by other Reddit users. Each subreddit (community) is moderated and managed by designated subreddit moderators who can also assign flags to specific comments. Authors who display inappropriate behavior can also be banned from specific subreddits (communities).

For the analysis discussed in this section, we extracted this dataset from Reddit using PRAW (Python Reddit API Wrapper), which is equipped with Python packages that provide access to Reddit's API. Using PRAW, we extracted 80938 distinct Reddit post data from 2020 to 2023. Moreover, we have extracted and mined the posting behavior of 59352 distinct Authors that posted their posts in at least one of the 103 different communities (subreddits) for the purpose of building a bipartite signed network. The node set of the signed bipartite network of the Reddit post dataset constitutes two types of nodes belonging to either the community (subreddit) set or the author set. Additionally, the edge set consists of edges

from authors to communities representing/recording the posting behavior of a given author in diverse communities.

The edges of the signed bipartite Reddit posts network represent that the given author (redditor) has posted a post in a given community (subreddit). These edges are further attributed with the primary edge features (refer Figure 1) such as Score, Upvote Ratio, Created On, Post URL, Distinguished, Edited, Over 18, Locked, Original Content, Total Comments, Author's name. The description of these primary edge features is discussed in Table 1. Furthermore, the attribute Upvote Ratio is utilized for assigning sign +1 or -1 to the edges of the resultant network. Meanwhile, the edges (posts) with an Upvote Ratio more than 0.5 are designated with sign +1, and the edges (posts) with an Upvote Ratio less than or equal to 0.5 are designated with sign -1. The Reddit posts network consists of a total of 80938 distinct edges (posts) among authors (redditors) and communities representing a signed relation on the basis of the upvote ratio of the post.

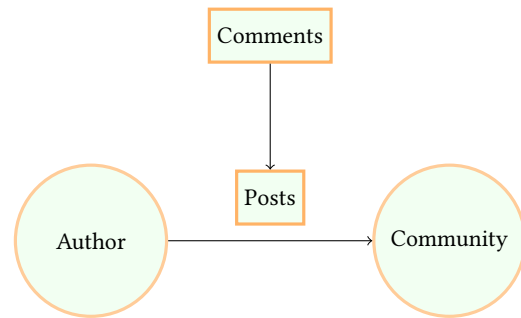


Figure 1: Diverse attributes with node level features (represented in circles) and edge level features (represented in rectangles) of Reddit Posts Dataset.

The Reddit post dataset consists of two types of nodes, communities and authors, that are also attributed to the node features. The nodes belonging to the community set are attributed with features such as ID, Sub_Name, Full_Name, Subscribers, Description, Created On, over18, Spoilers_enabled. The description of these node features is given in Table 2. Meanwhile, the nodes belonging to the author set are attributed with features such as ID, Name, Karma, Created On, Moderator, Trophies, Employee, Friend, and Premium User. The description of these node features is summarized in Table 3.

The posts posted by authors in a community are further discussed by other users in the form of comments and replies to those comments (comment forest). The PRAW specifically provides classes

¹<https://www.reddit.com/>

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Primary Features	Description
Subreddit ID	ID of the subreddit of the comment.
Score	The number of upvotes for the comment.
Upvote Ratio	The percentage of upvotes on the post.
Created On	Unix Time the post was created.
Post URL	The URL the post.
Distinguished	Whether or not the post is distinguished.
Edited	Whether or not the post has been edited.
Over 18	Whether or not the post has been marked as NSFW.
Locked	Whether or not the post has been locked.
Original Content	Whether or not the post has been set as original content.
Total Comments	number of comments on the post.
Author name	name of the submitting author of the post.

Table 1: Primary edge level features and their description.

Node Features	Description
ID	ID of the subreddit.
Sub_Name	Name of the subreddit.
Full_Name	Fullname of the subreddit.
Subscribers	Count of subscribers.
Description	Description of the subreddit, shown in searches.
Created On	Unix Time when the subreddit was created.
Spoilers_enabled	Whether the spoiler tag feature is enabled.
over18	Whether or not the subreddit has been marked as NSFW.

Table 2: Node level features of diverse communities (subreddits) and their description

Node Features	Description
ID	ID of the redditor.
Name	Username of the redditor.
Karma	The link karma for the redditor.
Created On	Unix Time when the account was created.
Trophies	list of the redditor's trophies.
Moderator	list of subreddits moderated and managed by the redditor.
Employee	Whether or not the redditor is a Reddit employee.
Friend	Whether or not the redditor is friends with the authenticated user.
Premium_User	Whether or not the redditor has active Reddit Premium status.

Table 3: Node level features of diverse Authors (redditors) and their description

such as Comment and CommentForest to extract these comments on various posts. We designate these attributes of the comments of a given post as secondary attributes (refer Figure 1) of the posts (edges). Hence, the edges (posts) are further attributed with secondary features such as Id, Author, Comment, Score, Created On, Replies, Submission, Subreddit, Subreddit_ID, Edited, Is_submitter, link_id, parent_id, Permalink. The attribute Upvote Ratio is not available for Comment and CommentForest classes using PRAW. The description of these secondary level edge features is given in Table 4.

1.2 Bitcoin Network Datasets

Proliferation in the use of online transactions and cryptocurrency has emerged as a transformation of the global financial landscape, reshaping traditional paradigms of commerce and payment systems. There are diverse cryptocurrencies available for anonymous online transactions, such as Bitcoin, Ethereum, Ripple, and Litecoin. Out of various available cryptocurrencies, Bitcoin is widely

Secondary Features	Description
ID	ID of the comment.
Author	Name of owner of the comment.
Comment	The body of the comment, as Markdown.
Score	The number of upvotes for the comment.
Created On	Unix time when the comment was created.
Replies	Provides an instance of replies of the comment.
Submission	The submission (post ID) that the comment belongs to.
Subreddit	The subreddit that the comment belongs to.
Subreddit_ID	The subreddit ID that the comment belongs to.
Edited	Whether or not the comment has been edited.
Is_submitter	Whether or not the comment author is also the author of the post.
link_id	The post ID that the comment belongs to.
parent_id	The ID of the parent comment (prefixed with t1_).
Permalink	A permalink for the comment.

Table 4: Secondary level edge features and their description.

considered the most famous and well-known cryptocurrency. Bitcoin is elementally employed for anonymous online transactions. The inherent anonymity also introduces complementary risk [2], such as those associated with over-the-counter or off-exchange trading platforms where two parties directly interact without the supervision of an exchange. This has therefore prompted the online community to establish various platforms where they can all publicly assess the trustworthiness of their counterparts to hopefully reduce the impact of potential scammers.

In this section, we discuss two Bitcoin web of trust network datasets: (1) Bitcoin-OTC¹ and (2) Bitcoin-Alpha². The Bitcoin-OTC dataset is extracted from its official website, where the dataset is available as the web of trust among its users. The Bitcoin-OTC platform allows its users to rate other users ranging from +10 to -10 (0 excluded). According to the guidelines of the platform, +10 designates the highest weighted trust, representing that the user is not a fraudster, whereas -10 designates the lowest weighted trust or total distrust, representing that the user is a fraudster. The other intermediate values among +10 and -10 hold the subsequent intermediate meanings.

We have extracted the web of trust among the users of the Bitcoin-OTC platform from 2010 to 2023 that consists of 5927 nodes and 36029 edges representing the weight of trust among two users based on their previous transactions (refer Figure 2). Moreover, the weighted edges of the Bitcoin-OTC signed network are supported by the timestamp and comment attributes. The timestamp attribute is a required field in those applications where the edge insertion and deletion depend on the time, for instance, network diffusion and modeling[1, 3]. Additionally, the comment attribute provides extra information in terms of validation of the weight rating by a user representing the trust for another user. Various other attributes of the Bitcoin-OTC signed network are as follows- RATER_User, RATED_User, RATING_TIME, Rating, Comment. The description of these attributes is given in Table 5.

The users of the Bitcoin-OTC signed network have the following node level attributes: User, REGISTERED_TIME, KEYID, TOTAL_RATING, NUM_POS_RATINGS_RECV, NUM_NEG_RATINGS_RECV, NUM_POS_RATINGS_SNT, NUM_NEG_RATINGS_SNT. the description of these attributes is given in Table 6. According

¹<https://bitcoin-otc.com/trust.php>²<https://btc-alpha.com/en>

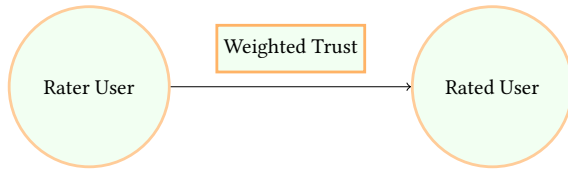


Figure 2: The structure of the Bitcoin network Datasets

Attributes	Description
RATER_User	ID of the Rater User.
RATED_User	ID of the Rated User.
RATING_TIME	Timestamp when the user was rated.
Rating	Weighted trust rating of the rated user.
Comment	Comment supporting the rating of the user.

Table 5: Various attributes of Bitcoin network Datasets and their description

to the data extracted, the earliest users registered on Bitcoin-OTC platform are from 2010 and the latest node is from 2023.

Node Features	Description
User	Name of the user of Bitcoin-OTC platform.
REGISTERED_TIME	Time when the user was registered
KEYID	Unique Hexadecimal Key.
TOTAL_RATING	Total rating of the user.
NUM_POS_RATINGS_RECV	Total positive rating received by the user.
NUM_NEG_RATINGS_RECV	Total negative rating received by the user.
NUM_POS_RATINGS_SNT	Total positive rating sent by the user.
NUM_NEG_RATINGS_SNT	Total negative rating sent by the user.

Table 6: Various node level features of Bitcoin-OTC signed network dataset and their description.

Similar to the Bitcoin-OTC platform, there is another platform for trading bitcoins known as Bitcoin-Alpha. We have extracted a signed dataset from Bitcoin-Alpha as a web of trust among its users. As in the case of Bitcoin-OTC, Bitcoin-Alpha also allows its users to rate other users based on their experience of previous transactions, ranging from +10 (maximum trust) to -10 (minimum trust), excluding 0. The extracted Bitcoin-Alpha signed dataset consists of a total of 3755 nodes and 24207 edges representing the weighted trust among two users of the Bitcoin-Alpha platform (refer Figure 2). These weighted edges are also attributed with features such as timestamps and comments. The attributes and structure of the Bitcoin-Alpha dataset are similar to the Bitcoin-OTC (refer Table 5 for attributes).

Additionally, the nodes of Bitcoin-Alpha dataset are attributed with node level features such as User, REGISTERED_TIME, LAST_SEEN_TIME, FINGERPRINT, AVG_RATING, TOTAL_RATING.

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Node Features	Description
User	Name of the user of Bitcoin-Alpha platform.
REGISTERED_TIME	Time when the user was registered
LAST_SEEN_TIME	Time when the user has last involved in any activity.
FINGERPRINT	Digital fingerprint of the user.
AVG_RATING	Average rating of the user.
TOTAL_RATING	Total rating of the user.

Table 7: Various node level features of Bitcoin-Alpha signed network dataset and their description.

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